

On Krull's Dimension Theorem for

1/13

Certain Graded Rings and Its Applications.

arXiv : 2605.00463

池田 吏来 (東京科学大 M2)

§ 1. Introduction

§ 2. Hilbert — Serre Rings

§ 3. Main Theorem

§ 4. Examples and Question.

Notation

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

$$R = \bigoplus_{n \in \mathbb{N}} R_n \quad : \quad \mathbb{N}\text{-graded} \quad (R_0 = \mathbb{K} : \text{field})$$

$$P_R(t) = \sum_{n \in \mathbb{N}} (\dim_{\mathbb{K}} R_n) t^n \quad : \quad \text{Poincaré series.}$$

§ 1. Introduction

3/13

$$\circ \operatorname{clig}_R R := \sup \{ r \in \mathbb{N} \mid P_0 \subsetneq \dots \subsetneq P_r : \text{homog prime chain of } R \}$$

$$\circ \operatorname{cl}_i R := \text{krull dimension}$$

$$\circ \operatorname{tr. deg}_k R := \text{transcendence degree of } R \text{ over } k$$

$$\circ \operatorname{GK dim}_k R := \sup_V \limsup_{n \rightarrow \infty} \frac{\log \operatorname{clig}_k (k + V + \dots + V^n)}{\log n}$$

$$\circ d(R) := \min \{ d \in \mathbb{N} \mid \lim_{t \rightarrow 1-0} (1-t)^d P_R(t) < \infty \}$$

Thm 1.1 (Serre's Dimension Theorem)

4/13

R : Noether $\mathbb{N}$ -graded ring

$$\dim R = \mathcal{A}(R) = s(R)$$

□

$s(R) : \min \{ s \in \mathbb{N} \mid \exists x_1, \dots, x_s \in R : \text{homog s.t.}$

$\varphi[x_1, \dots, x_s] \rightarrow R : \text{module fin } \}$

Def 1.2 (Initial Algebras)

5/13

$$R = k[f_1, \dots, f_m] \subset k[x_1, \dots, x_n]$$

($f_1, \dots, f_m$: homog)

$<$: monomial order

$$\text{in}_<(R) := k[\text{in}_<(f) \mid f \in R]$$

□

⌈ monomial algebra

If $\text{in}_<(R)$ is Noeth, we have

$$\text{dim } R = d(R) = d(\text{in}_<(R)) = \text{dim}(\text{in}_<(R))$$

↑ ↑ ↑
Smoke Poincaré
 series
 coincide

$\text{in}_<(R)$ is not necessarily Noeth in general!

$$\left| \frac{6}{13} \right|$$

Example 1.3

$$R = k[x+y, xy, xy^2] \subset k[x, y].$$

$\text{in}_<(R)$ is not Noeth for all monomial orders.

□

§ 2 Hilbert - Serre Rings

7/13

Def 2.1 (Hilbert - Serre rings)

R : Hilbert - Serre ring $\Leftrightarrow$
def

$$(1) \dim_k R_n < \infty \quad (\forall n \in \mathbb{N})$$

(2) $P_R(t)$: radius of convergence 1 or ∞ .

$$(3) D = \{ d \in \mathbb{N} \mid \lim_{t \rightarrow 1-0} (1-t)^d P_R(t) < \infty \} \neq \emptyset.$$

□

$d(R) = \min D$: Hilbert - Serre dimension.

Example 2.2

$$\boxed{8/13}$$

$$(1) R: \text{Noeth} \Rightarrow R: \text{Hilbert-Serre}$$

$$(2) R = \mathbb{C}[x, xy, xy^2, xy^3, \dots] \subset \mathbb{C}[x, y] : \text{Hilbert-Serre}$$
$$\mathcal{A}(R) = 2.$$

$$(3) S = \bigoplus_{n \in \mathbb{N}} S_n \hookrightarrow \bigoplus_{n \in \mathbb{N}} R_n : \text{homog ring hom} \quad (S_0 = \mathbb{C})$$

$$R: \text{Hilbert-Serre} \Rightarrow S: \text{Hilbert-Serre} \quad \text{and} \quad \mathcal{A}(S) \leq \mathcal{A}(R)$$

□

§ 3 Main Theorem

9/13

Thm 3.1 R : Hilbert - Serre ring

$$\text{clgr } R \leq \text{cl } R \leq \text{GKdim } R \leq d(R)$$

If R : domain

$$\text{clgr } R \leq \text{cl } R \leq \text{tr.deg } R = \text{GKdim } R \leq d(R)$$

If R : monomial algebra,

$$\text{clgr } R = \text{cl } R = \text{tr.deg } R = \text{GKdim } R = d(R)$$

□

$$\left\lfloor \frac{10}{13} \right\rfloor$$

$$R = k[t_1, \dots, t_m] \subset k[x_1, \dots, x_n]$$

$<$: mon order

$$\dim R = d(R) = d(\text{in}_<(R)) = \dim \text{in}_<(R)$$

§ 4. Examples and Question

11/13

$r, c \in \mathbb{N}_{>0}$ k : countable alg clsd field

$$(k = \overline{\mathbb{Q}}, \overline{\mathbb{F}}_p)$$

$$X = \mathbb{P}_{k^r}^r = \text{Proj}(k[x_0, x_1, \dots, x_r])$$

$K = k(x_1, \dots, x_r)$: rational func field of X .

$C_0 = \{x_0=0\}$, $C_1 = \{x_1=0\}$, $C_2, C_3, \dots$: enumerating
all prime divisors

$$E_{-1}, E_0 = 0, \quad E_n = n^c C_0 + \sum_{i=1}^{\infty} \left\lfloor \frac{n}{2^i \deg C_i} \right\rfloor C_i$$

$$\left\lfloor \frac{12}{13} \right\rfloor$$

$$R = \bigoplus_{n \in \mathbb{N}} H^0(X, \mathcal{O}_X(E_{n-1})) T^n \subset k(x_0, \dots, x_r)[T]$$

$$\begin{array}{ccccccc} \dim_{gr} R & < & \dim R & \leq & \text{tr.deg}_k R & \leq & \mathcal{A}(R) \\ \parallel & & \parallel & & \parallel & & \parallel \\ 1 & & 2 & & r+1 & & cr+1 \end{array}$$

Question, $a, b, c, d, e \in \mathbb{N}$

13/13

$\exists R$: Hilbert-Serre domain s.t.

$$\dim_{\mathbb{Q}} R = a, \dim R = b$$

$$\text{tr. deg}_k R = c, \text{GKdim}_k R = d, \dim(R) = e \quad ? \quad \square$$

necessary conditions

$$\circ a \leq b \leq c = d \leq e$$

$$\circ \left\lfloor \frac{b}{2} \right\rfloor \leq a$$